



Identifying a Research Problem

PPT PRESENTED BY

NAME : Srinivas Rao Tammireddy

ROLL NO : 257Y1D5805

YEAR : I Year I Semester

ACADEMIC YEAR : 2025 - 2026

What is a Research Problem?

A Research Problem is a clear, precise expression of an area of concern, a gap or contradiction in literature, or a practical challenge that requires systematic investigation to solve.

Characteristics of a Good Problem

- **Novelty & Originality:** Must address a genuine gap. Should not duplicate completed research.
- **Feasibility & Resources:** Can be solved within standard constraints (time, budget, equipment).
- **Social & Scientific Value:** Practical or theoretical relevance, adding value to academia or industry.
- **Scope Boundary:** Clearly bounded so it is manageable — avoiding topics that are too broad.

Primary Sources of Research Problems

- **Literature Gaps:** Contradictory findings in journals, recommendations for future study.
- **Practical Challenges:** Real-world engineering issues, system inefficiencies, or software bugs.
- **Technological Changes:** New tools (AI, cloud) create fresh domains needing new research.
- **Conceptual Theories:** Testing established scientific theories in new, unexplored contexts.

Errors in Problem Selection

Researchers often commit classic errors when selecting and defining their research problem:

Common Selection Errors

- **Choosing Too Broad a Topic:** E.g., 'Improving Cloud Database Uptime'. A single research paper cannot solve this. It must be narrowed to a specific query, layer, or mechanism.
- **Choosing Too Narrow a Topic:** E.g., 'Optimizing database index lookups for one local restaurant system'. Lacks general applicability or scientific value.
- **Overlooking Constraints:** Selecting problems requiring equipment (like supercomputers or specialized hardware) the university does not possess.
- **Pre-conceived Bias:** Selecting a topic to prove a personal bias rather than exploring objectively.

Approaches of Investigation

- **Systematic Data Collection:** Formulating protocols to log experimental parameters or user metrics.
- **Analytical Interpretation:** Applying statistical validation tests (t-tests, ANOVA) to verify if results are significant.
- **Instrumentation Requirements:** Identifying the necessary software tools, datasets, and hardware metrics (e.g. CPU temperature, network latency) to measure results accurately.

Systematic Investigation Flow

The scientific process of investigating a research problem proceeds in six steps:

The Six Steps of Investigation

- 1. Identify & Define the Problem:** Translate an area of concern into a specific, bounded research question.
- 2. Construct Hypotheses:** Formulate testable assertions about relationships between variables.
- 3. Design Experiments:** Plan the test setup, control conditions, and target metrics.
- 4. Collect Data:** Run code simulations, query database testbeds, or gather hardware logs systematically.
- 5. Analyze & Interpret:** Apply statistical modeling (e.g., regressions) to process values.
- 6. Draw Conclusions & Publish:** Determine if the hypotheses were validated. Document limitations and publish findings.

Applications of Problem Formulation

Formulating a precise research problem is critical in academic and corporate domains:

Research Grant Proposals

Funding bodies (e.g., DST, UGC, corporate foundations) reject proposals that have vague problem definitions. A clear, well-formulated problem statement is essential for securing research funding.

Master's & PhD Thesis

Students must defend their thesis topic choice before review committees. A clearly bounded, novel problem statement guarantees a clear direction for the research.

Corporate R&D Initiatives

Tech companies frame R&D goals as precise problems (e.g., 'Reducing latency in distributed databases under 90% load'). Guides engineering teams towards clear performance targets.

Standardizing Patents

Patent offices require inventions to solve a specific 'technical problem' in a non-obvious way. A precise definition of the problem is key to getting a patent granted.



THANK YOU

Department of Computer Science and Engineering
Marri Laxman Reddy Institute of Technology and Management